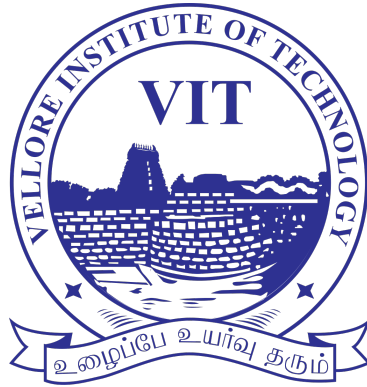


BookIt Resource Booking System

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Project Report



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November 24, 2025

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Chapter 1

Introduction

1.1 What is BookIt?

BookIt is a simple desktop application that helps students book study rooms, labs, and other resources at our college. Instead of running around campus trying to find a free room or waiting in long lines, you can now book everything from your computer!

1.2 Why did we build this?

We noticed that students often struggle to find available resources when they need them most. Sometimes you'd go to the library only to find all the study rooms taken. Other times, you'd need to use a special equipment but didn't know how to reserve it. BookIt solves these problems by putting everything in one place.

1.3 What can you do with it?

With BookIt, you can:

- See all available resources in college
- Check when they're free
- Book them with just a few clicks
- View and cancel your bookings

Chapter 2

Problem Statement

2.1 The Problem We Faced

Finding and booking resources in college was a headache! Here's what students were dealing with:

- **No idea what's available:** You had to physically go to check if a room was free
- **Wasted time:** Walking around campus only to find everything booked
- **Confusion:** Different departments had different booking systems
- **Double bookings:** Sometimes two people would book the same room
- **Forgotten bookings:** No reminders about your upcoming bookings

2.2 How it affected everyone

- Students couldn't plan their study sessions properly
- Teachers couldn't book resources for their classes efficiently
- College resources weren't being used optimally
- Lots of arguments over who booked what first

2.3 Our Solution

BookIt makes everything simple:

- See everything available in one place
- Book from anywhere (even from home!)
- No more double bookings
- Easy to manage your bookings

Chapter 3

Functional Requirements

3.1 What the App Must Do

3.1.1 Login System

- Users need to login with username and password
- Two types of users: students and admins
- Different screens for different users

3.1.2 For Students

- Browse all resources (rooms, labs, equipment)
- See details like location, capacity, etc.
- Book resources for specific dates and times
- Check if the resource is available before booking
- View all your bookings
- Cancel bookings if plans change

3.1.3 For Admins

- Add new resources to the system
- See all bookings made by everyone
- View statistics (how many bookings, which resources are popular)

3.1.4 General Features

- Prevent double bookings
- Show helpful messages (success, errors, etc.)
- Logout when done

Chapter 4

Non-functional Requirements

4.1 How the App Should Work

4.1.1 Easy to Use

- Simple, clean interface
- Clear instructions
- Same look and feel throughout

4.1.2 Fast and Responsive

- Quick loading
- No delays when booking
- Works smoothly even with many users

4.1.3 Reliable

- Doesn't crash or lose data
- Handles errors gracefully
- Keeps bookings safe

4.1.4 Secure

- Only logged-in users can book
- Students can't see admin features
- Passwords are protected

Chapter 5

System Architecture

5.1 How We Built It

5.1.1 The Big Picture

Think of BookIt like a building with different floors:

- **Ground Floor (Login):** Where everyone enters
- **First Floor (Student Area):** Where students do their booking
- **Second Floor (Admin Area):** Where admins manage everything
- **Basement (Data Storage):** Where all information is kept

5.1.2 Main Parts

- **Login Screen:** Checks who you are
- **Student Dashboard:** Your personal booking space
- **Admin Dashboard:** Management area
- **Booking Engine:** The brain that handles all bookings
- **Data Storage:** Memory that remembers everything

5.1.3 Technologies Used

- **Python:** The programming language
- **Tkinter:** For creating the windows and buttons
- **Lists and Dictionaries:** For storing data (like a digital filing cabinet)

Chapter 6

Design Diagrams

6.0.1 User Flow

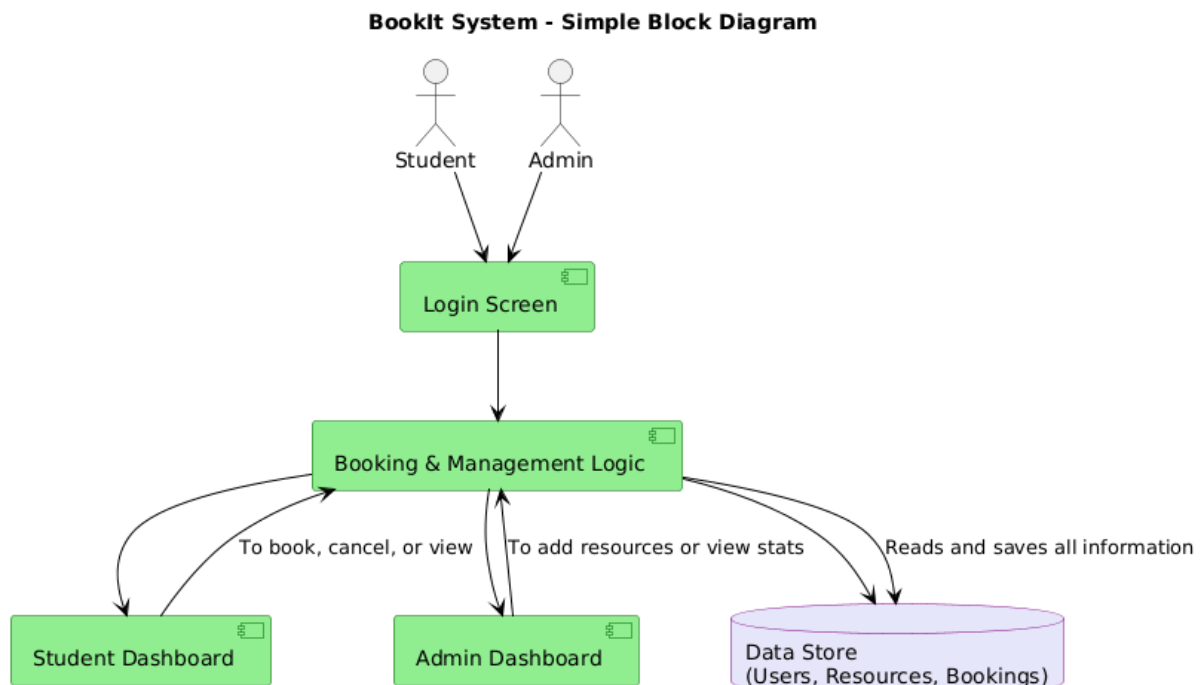


Figure 6.1: Block Diagram of system

Chapter 7

Design Decisions & Rationale

7.1 Why We Made These Choices

7.1.1 Technology Choices

- **Python + Tkinter:** Easy to learn, works on all computers, perfect for beginners
- **Simple Data Storage:** Used lists instead of a database to keep things simple for the first version

7.1.2 Design Choices

- **Tabs instead of multiple windows:** Cleaner and less confusing
- **Color coding:** Green for success, red for errors - easy to understand
- **Simple forms:** Just fill in what's needed, nothing extra

7.1.3 Why Not Something Else?

- Could have used a database, but that would make it complicated
- Could have made it web-based, but desktop apps are simpler to start with
- Could have added more features, but we wanted to get the basics right first

Chapter 8

Implementation Details

8.1 How We Coded It

8.1.1 The Login System

Simple username and password check against a list of users. If it matches, you're in! If not, try again.

8.1.2 Student Dashboard

Three tabs to keep things organized:

- **Browse:** See everything available
- **Book:** Make new bookings
- **My Bookings:** See and cancel your bookings

8.1.3 Admin Dashboard

Three tabs for management:

- **Add Resource:** Add new rooms or equipment
- **All Bookings:** See what everyone has booked
- **Stats:** Numbers and charts about usage

8.1.4 Important Functions

- `handle_booking_confirmation()`: Checks if booking is okay and saves it
- `handle_booking_cancellation()`: Removes bookings you don't want
- `handle_add_resource()`: Adds new resources to the system

8.1.5 Error Handling

What happens when things go wrong:

- Wrong password: Shows error message
- Invalid date/time: Tells you what's wrong
- Double booking: Prevents it and lets you know
- Empty fields: Reminds you to fill everything

Chapter 9

Testing Approach

9.1 How We Made Sure It Works

9.1.1 Testing Each Part

We tested every button and feature:

- Login with correct and wrong passwords
- Try to book the same time twice with missing information
- Add new resources
- Cancel bookings

9.1.2 Testing the Whole Flow

We tested complete scenarios:

- From login to successful booking
- From login to canceling a booking
- Admin adding a resource and student booking it

9.1.3 Testing with Real Users

We asked other students to try it:

- Was it easy to understand?
- Did they get stuck anywhere?
- Did they like using it?

9.2 Test Results

What We Tested	What Should Happen	What Actually Happened
Login with correct details	Go to dashboard	Worked perfectly!
Login with wrong details	Show error message	Worked perfectly!
Make a valid booking	Confirm and save	Worked perfectly!
Try double booking	Show conflict error	Worked perfectly!
Add new resource	Show in list	Worked perfectly!
Cancel booking	Remove from list	Worked perfectly!

Table 9.1: Our Testing Results

Chapter 10

Learnings & Key Takeaways

10.1 What We Learned

10.1.1 Technical Skills

- How to build desktop apps with Python
- Making buttons and windows that actually work
- Checking user input and preventing errors
- Organizing code so it doesn't become a mess

10.1.2 Design Skills

- Simple is better than complicated
- Colors and layout matter a lot
- Users need clear instructions
- Feedback (like success messages) is important

10.1.3 Project Skills

- Breaking big problems into small pieces
- Testing as you go (not just at the end)
- Writing comments in code (future you will thank you!)
- Planning before coding saves time

10.2 Challenges We Faced

- Keeping track of what's happening in different windows
- Making sure data doesn't get mixed up
- Making it look good on different screen sizes

- Deciding what features to include vs. what to leave out

10.3 Proud Moments

- When the first successful booking worked!
- When other students said it was easy to use
- When we fixed a really tricky bug
- When the admin dashboard actually showed real statistics

Chapter 11

Future Enhancements

11.1 What's Next for BookIt

11.1.1 Cool Features We Want to Add

- **Calendar View:** See bookings in a calendar format
- **Recurring Bookings:** Book the same slot every week
- **Waitlist:** Get notified when a busy resource becomes free
- **Ratings:** Rate resources after using them
- **Reminders:** Get email/SMS before your booking

11.1.2 Technical Improvements

- **Database:** Store data permanently (not just when running)
- **Web Version:** Use it on phones and tablets
- **Better Security:** Encrypted passwords and secure login
- **Reports:** Generate PDF reports for admins

11.1.3 User Experience Improvements

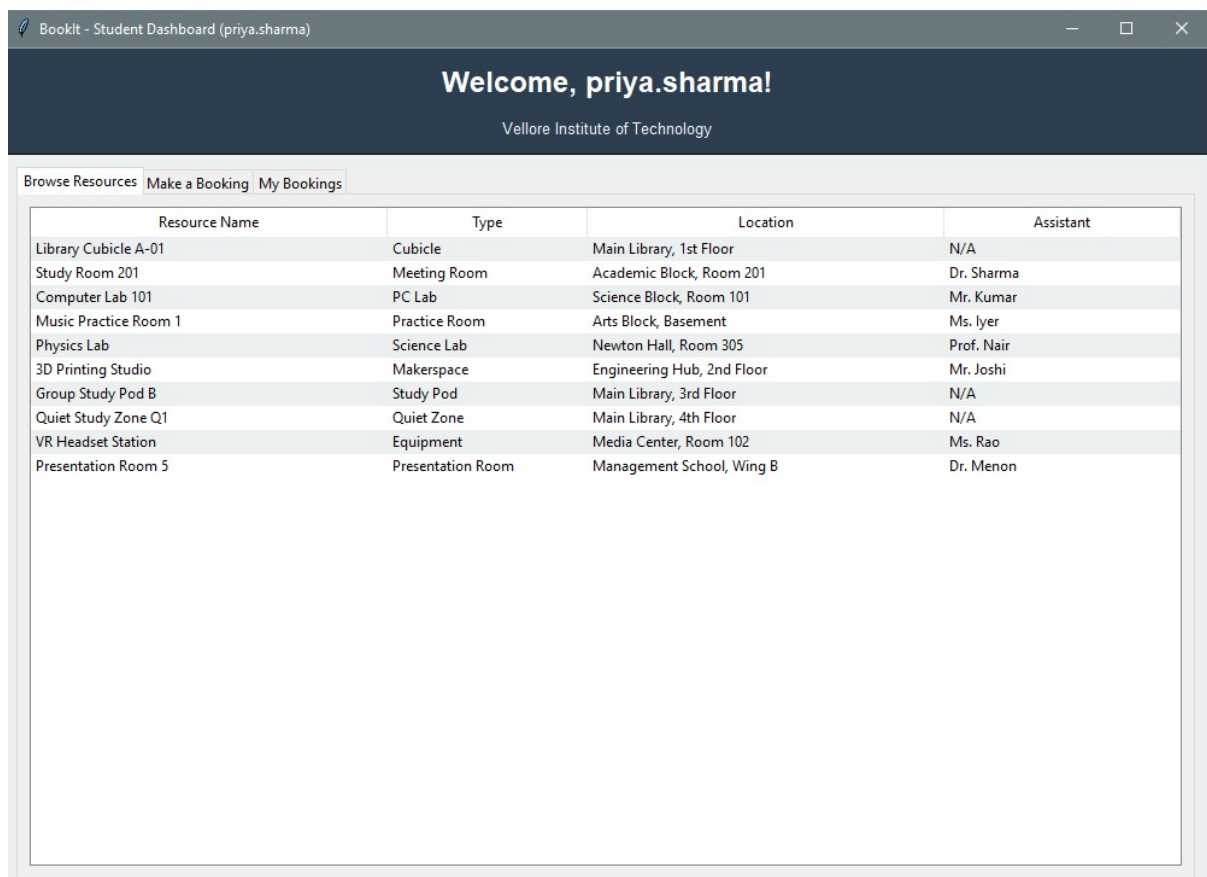
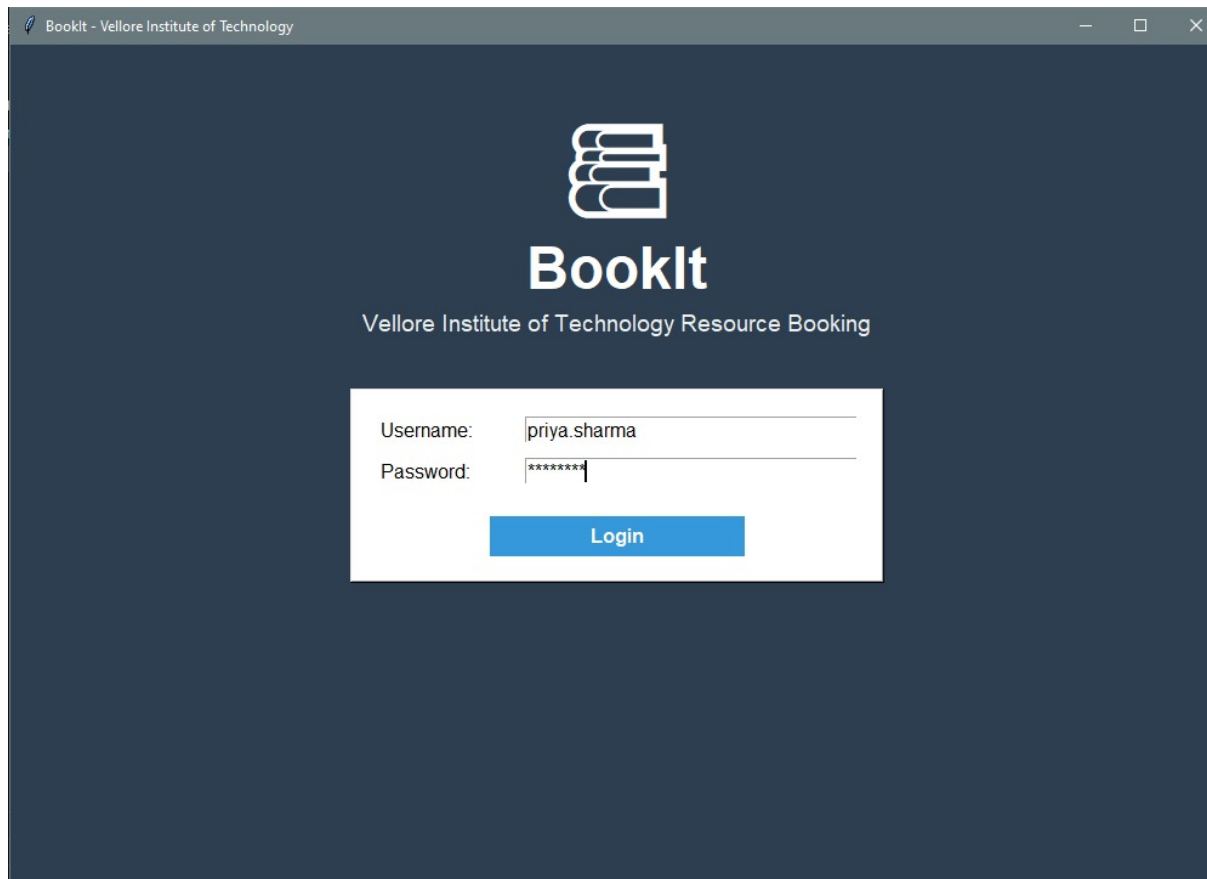
- **Search and Filter:** Find exactly what you need quickly
- **User Profiles:** Save your preferences and history
- **Photos:** See pictures of resources before booking
- **Dark Mode:** For late-night booking sessions!

11.1.4 Admin Features

- **User Management:** Add/remove students
- **Advanced Scheduling:** Set rules for bookings
- **Usage Analytics:** Detailed reports on resource usage
- **Maintenance Mode:** Temporarily close resources for repairs

11.2 Dream Features

- AI-powered suggestions based on your booking history
- Integration with college timetable
- Automatic room allocation based on group size
- Voice commands for booking
- Integration with college ID cards for automatic check-in



BookIt - Student Dashboard (priya.sharma)

Welcome, priya.sharma!
Vellore Institute of Technology

Browse Resources | Make a Booking | My Bookings

Select a Resource: 2: Study Room 201

Date (YYYY-MM-DD): 2: Study Room 201

Start Time (HH:MM):

- 1: Library Cubicle A-01
- 2: Study Room 201
- 3: Computer Lab 101
- 4: Music Practice Room 1
- 5: Physics Lab
- 6: 3D Printing Studio
- 7: Group Study Pod B
- 8: Quiet Study Zone Q1
- 9: VR Headset Station
- 10: Presentation Room 5

BookIt - Student Dashboard (priya.sharma)

Welcome, priya.sharma!
Vellore Institute of Technology

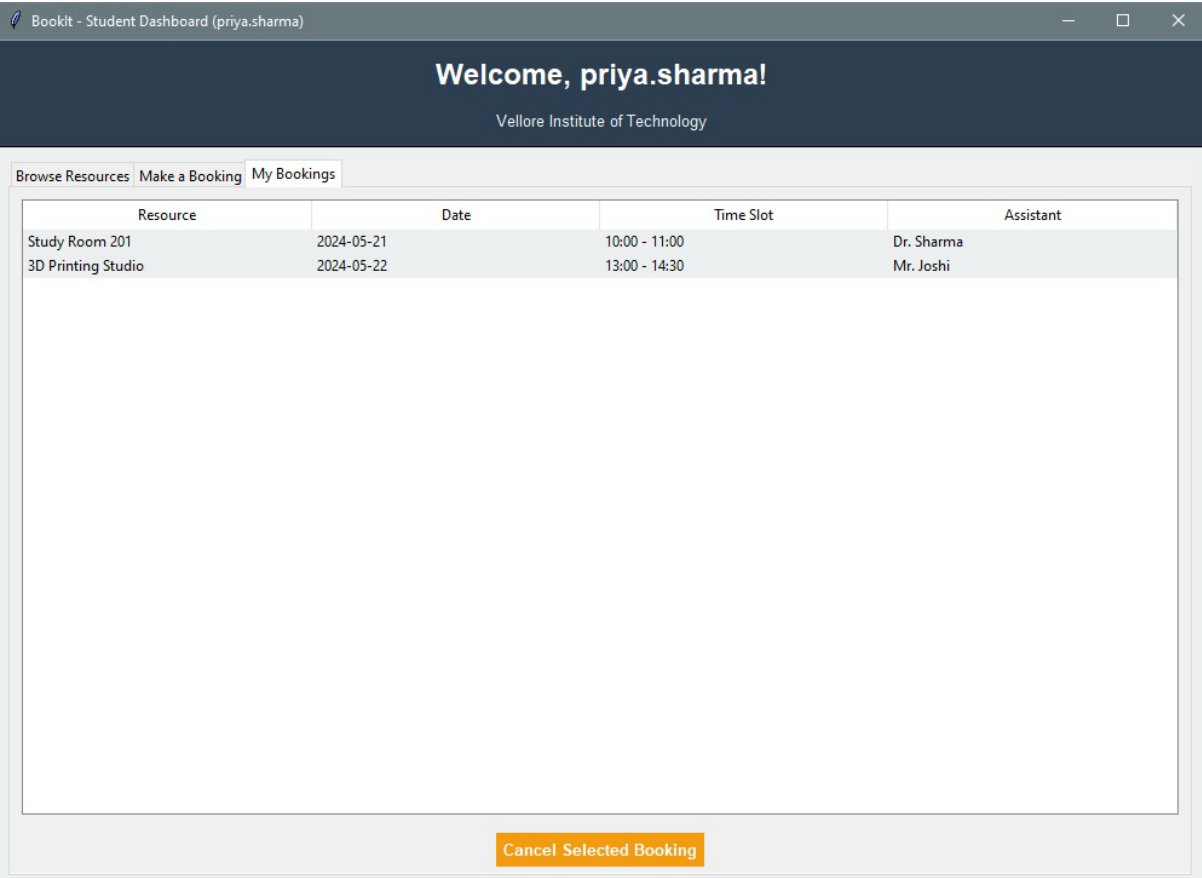
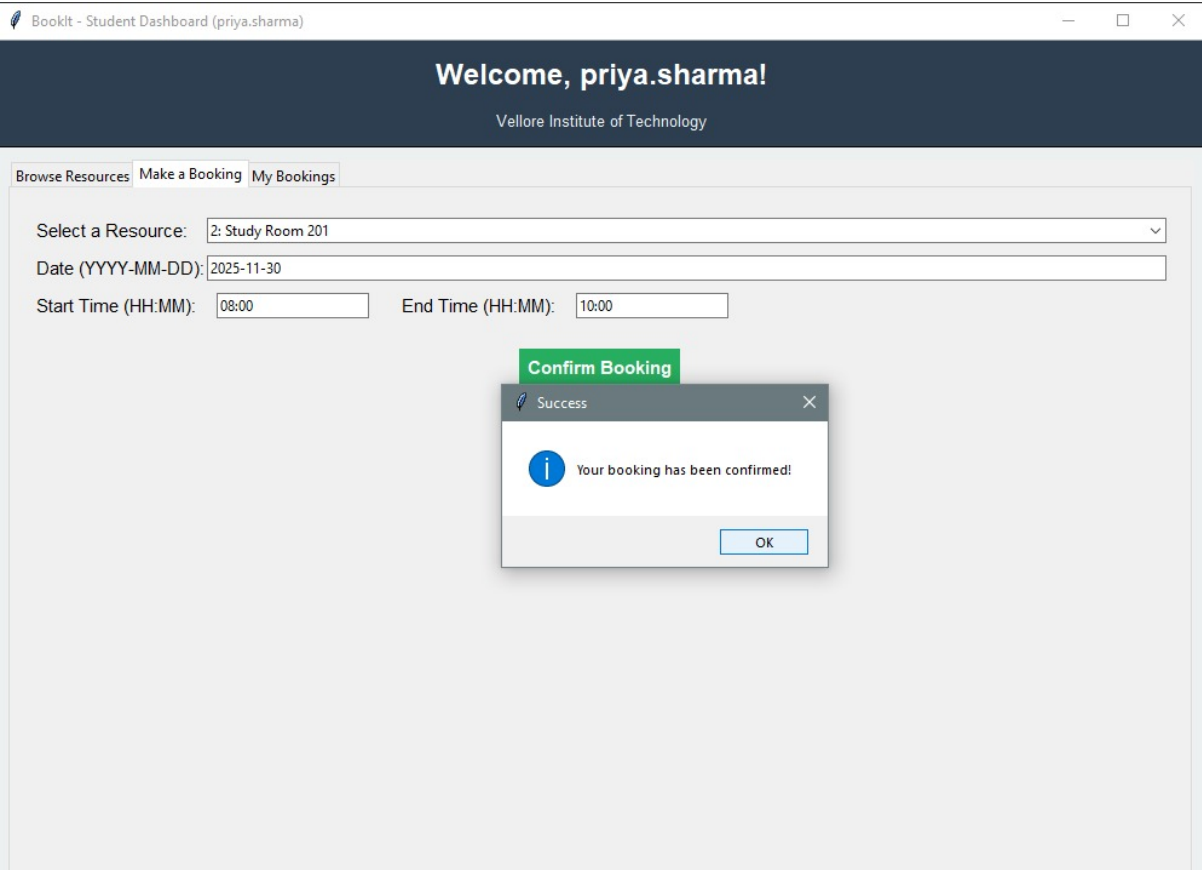
Browse Resources | Make a Booking | My Bookings

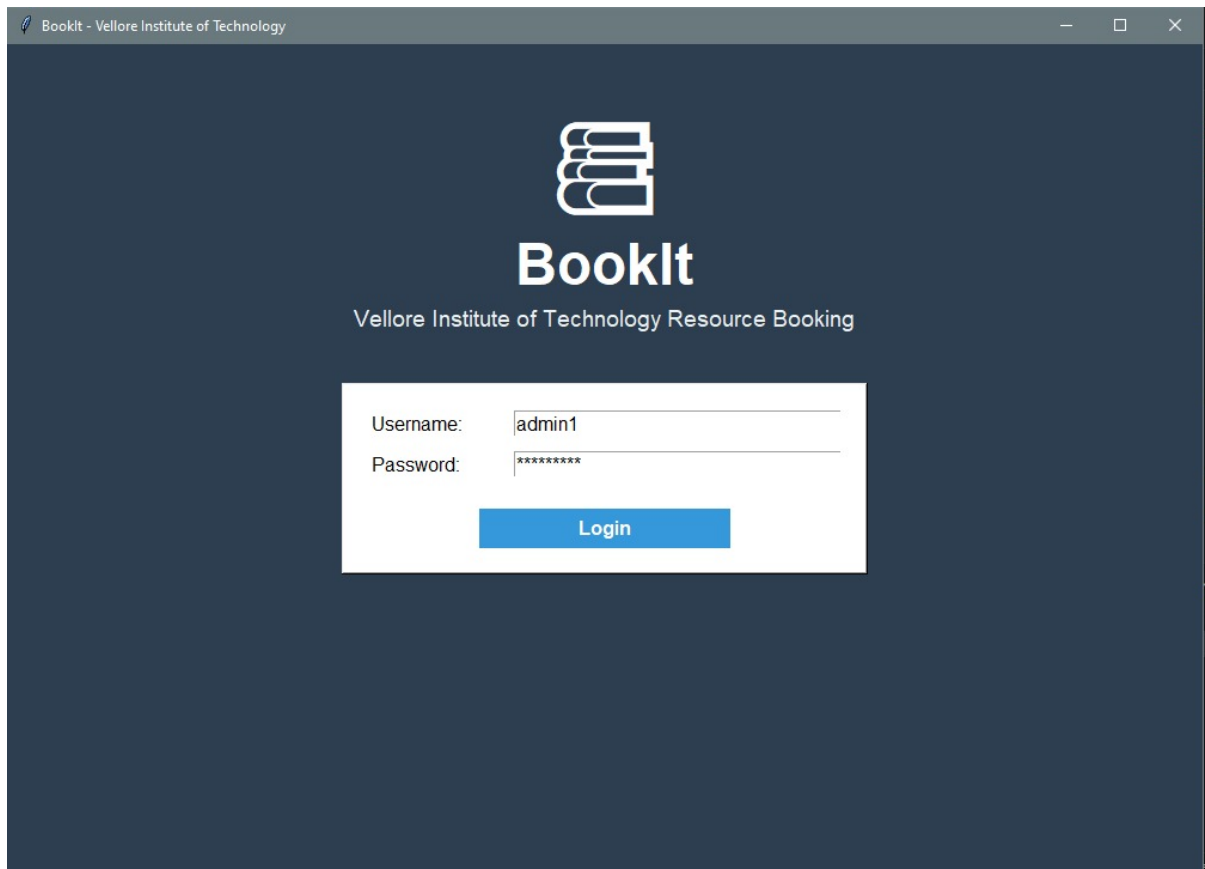
Select a Resource: 2: Study Room 201

Date (YYYY-MM-DD): 2025-11-30

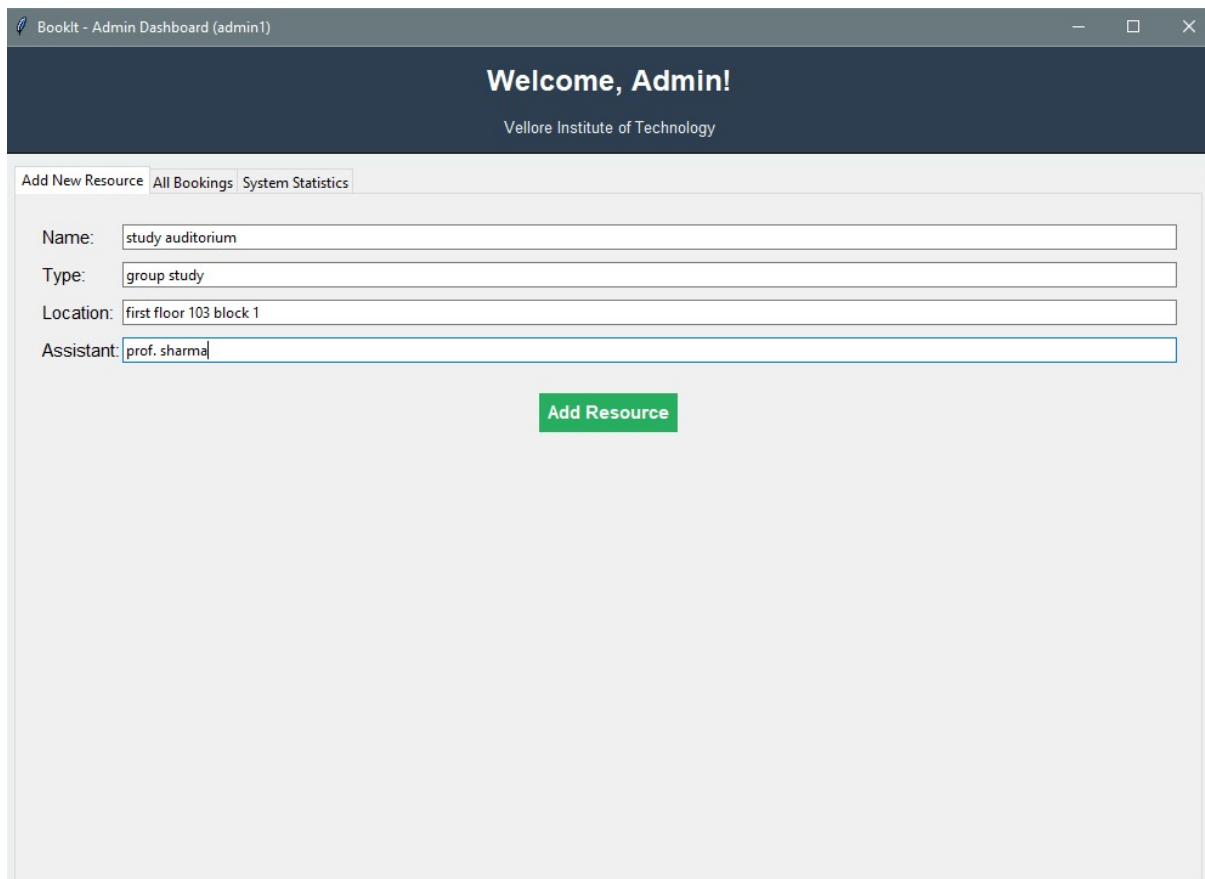
Start Time (HH:MM): 08:00 End Time (HH:MM): 10:00

Confirm Booking





The screenshot shows the login page of the BookIt system. The browser window title is "BookIt - Vellore Institute of Technology". The page has a dark blue background. At the top center is a logo consisting of three stacked books. Below the logo, the text "BookIt" is displayed in a large, white, sans-serif font. Underneath "BookIt" is the text "Vellore Institute of Technology Resource Booking" in a smaller, white, sans-serif font. In the center of the page is a white rectangular box containing the login form. The form has two input fields: "Username:" with the value "admin1" and "Password:" with the value "*****". Below these fields is a blue button labeled "Login".



The screenshot shows the Admin Dashboard of the BookIt system. The browser window title is "BookIt - Admin Dashboard (admin1)". The page has a dark blue header with the text "Welcome, Admin!" and "Vellore Institute of Technology" below it. Below the header is a light gray sidebar with three tabs: "Add New Resource", "All Bookings", and "System Statistics". The "Add New Resource" tab is selected. The main content area is light gray and contains a form with four input fields: "Name:" with the value "study auditorium", "Type:" with the value "group study", "Location:" with the value "first floor 103 block 1", and "Assistant:" with the value "prof. sharma". Below these fields is a green button labeled "Add Resource".

BookIt - Admin Dashboard (admin1)

Welcome, Admin!

Vellore Institute of Technology

Add New Resource

All Bookings

System Statistics

Student	Resource	Date	Time Slot	Assistant
priya.sharma	Study Room 201	2024-05-21	10:00 - 11:00	Dr. Sharma
rohan.verma	Computer Lab 101	2024-05-21	14:00 - 16:00	Mr. Kumar
priya.sharma	3D Printing Studio	2024-05-22	13:00 - 14:30	Mr. Joshi
ananya.iyer	Physics Lab	2024-05-23	09:00 - 12:00	Prof. Nair
vikram.patel	Presentation Room 5	2024-05-24	15:00 - 17:00	Dr. Menon

BookIt - Admin Dashboard (admin1)

Welcome, Admin!

Vellore Institute of Technology

Add New Resource

All Bookings

System Statistics

Student	Resource	Date	Time Slot	Assistant
priya.sharma	Study Room 201	2024-05-21	10:00 - 11:00	Dr. Sharma
rohan.verma	Computer Lab 101	2024-05-21	14:00 - 16:00	Mr. Kumar
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